Solar Energy Forecasting

Problem & Motivation

NEED

Accurate short-term solar forecasts are essential for ensuring **grid stability**. They help manage supply and demand effectively in energy systems.

CHALLENGES

Variability in irradiance and weather conditions presents significant challenges, complicating operational planning and making reliable forecasting crucial for energy providers.

OBJECTIVE

The primary objective of this project is to **predict solar power generation** 15 minutes ahead using machine learning techniques to enhance accuracy.

Dataset

SOLAR PLANT OPERATIONAL DATA

The dataset from Kaggle includes essential columns such as irradiation, ambient temperature, and generation, along with engineered features for improved forecasting.

DATASET LINK

GITHUB LINK

Data Preprocessing Steps

TIMESTAMP CONVERSION

Converting timestamps to datetime format allows for proper data manipulation and merging, ensuring alignment across various datasets for analysis.

MISSING VALUES

Handling missing values is crucial to maintain data integrity, employing methods like imputation to fill gaps and avoid biases in predictions.

FEATURE CREATION

Creating lag features and rolling statistics enhances the model's predictive capabilities, capturing trends and patterns over time for improved accuracy.

Model Training Overview

RANDOM FOREST REGRESSOR

We utilized a Random Forest Regressor model, which is robust and effective in handling non-linear relationships commonly found in solar data.

TRAINING AND TESTING

The dataset was split into training and testing sets to evaluate the model's performance effectively while optimizing for feature set selection.

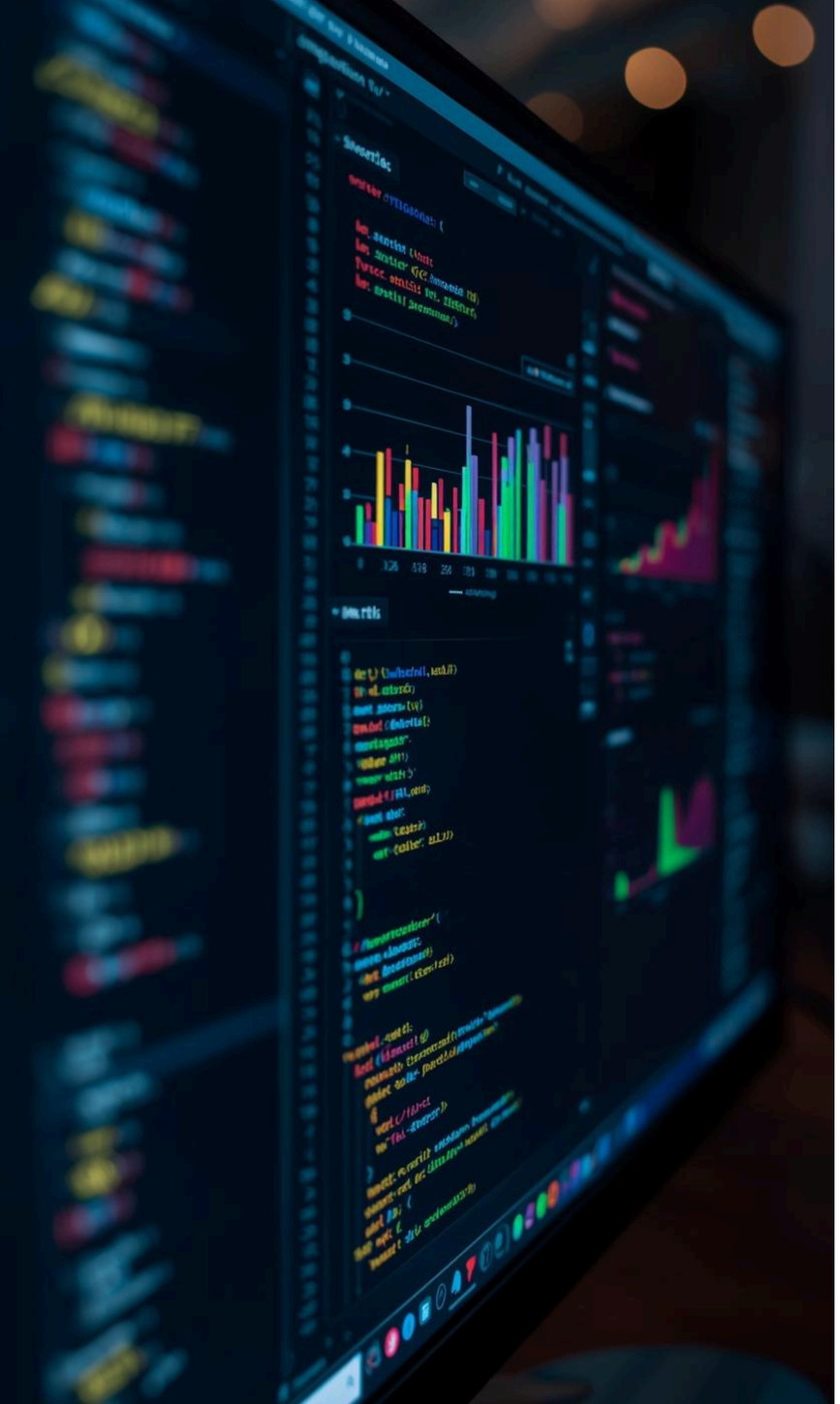
Evaluation Metrics

```python
model = RandomForestRegressor(
    n_estimators=200,
    max_depth=12,
    random_state=42,
    n_jobs=-1
)

model.fit(X_train, y_train)

preds = model.predict(X_test)

mae = mean_absolute_error(y_test, preds)
rmse = np.sqrt(mean_squared_error(y_test, preds))
r2 = r2_score(y_test, preds)

print("MAE:", mae)
print("RMSE:", rmse)
print("R2:", r2)
```

```
MAE: 10573.810102716383
RMSE: 21207.710057670523
R2: 0.9323337701435448
```

KEY PERFORMANCE INDICATORS

The primary evaluation metrics include **Mean Absolute Error (MAE)**, **Root Mean Squared Error (RMSE)**, and **R-squared (R2)**, which quantify model accuracy.



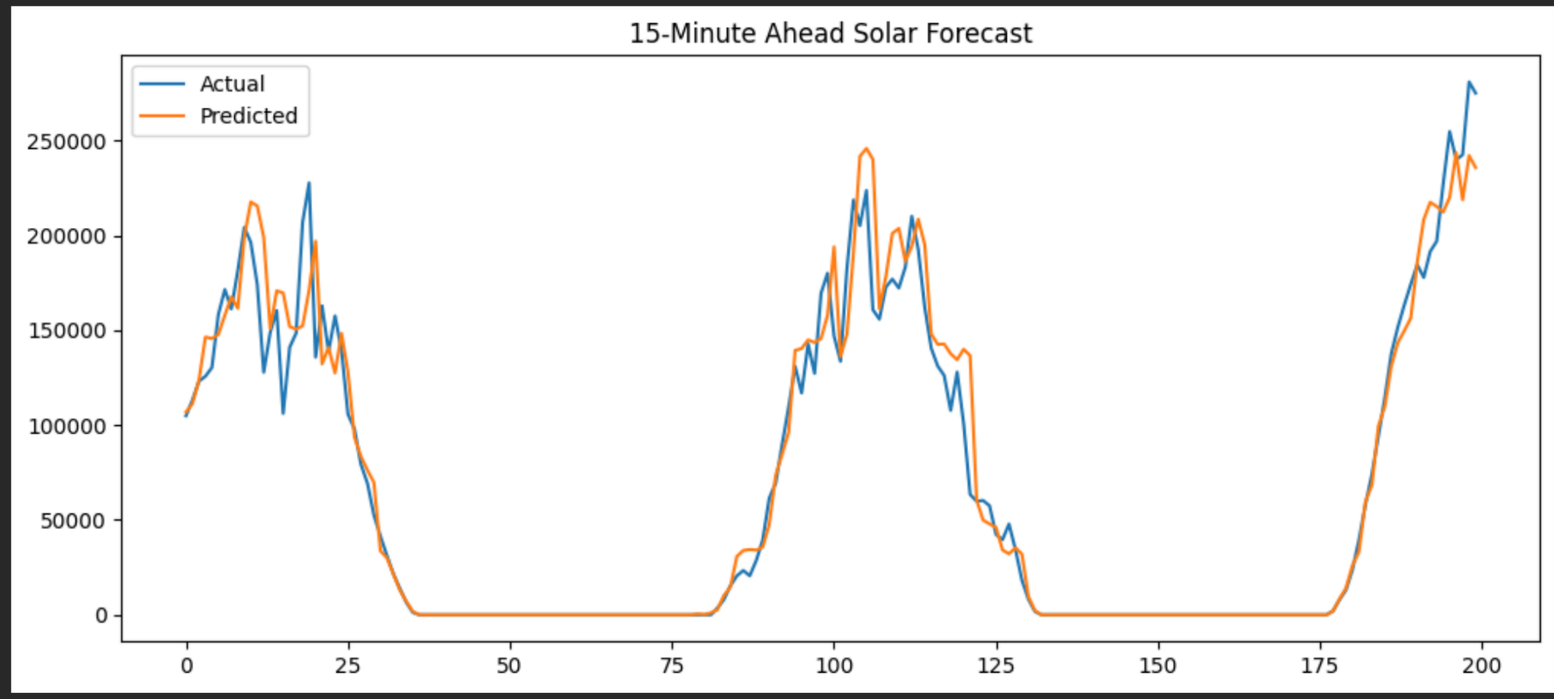
93%

MODEL ACCURACY

The model achieves an impressive **93% accuracy**, demonstrating its effectiveness in predicting solar energy generation. This high level of precision indicates that the machine learning approach successfully captures the underlying patterns and variability present in solar power data, making it a reliable tool for grid stability.

Accurate Solar Power Predictions in Real Time

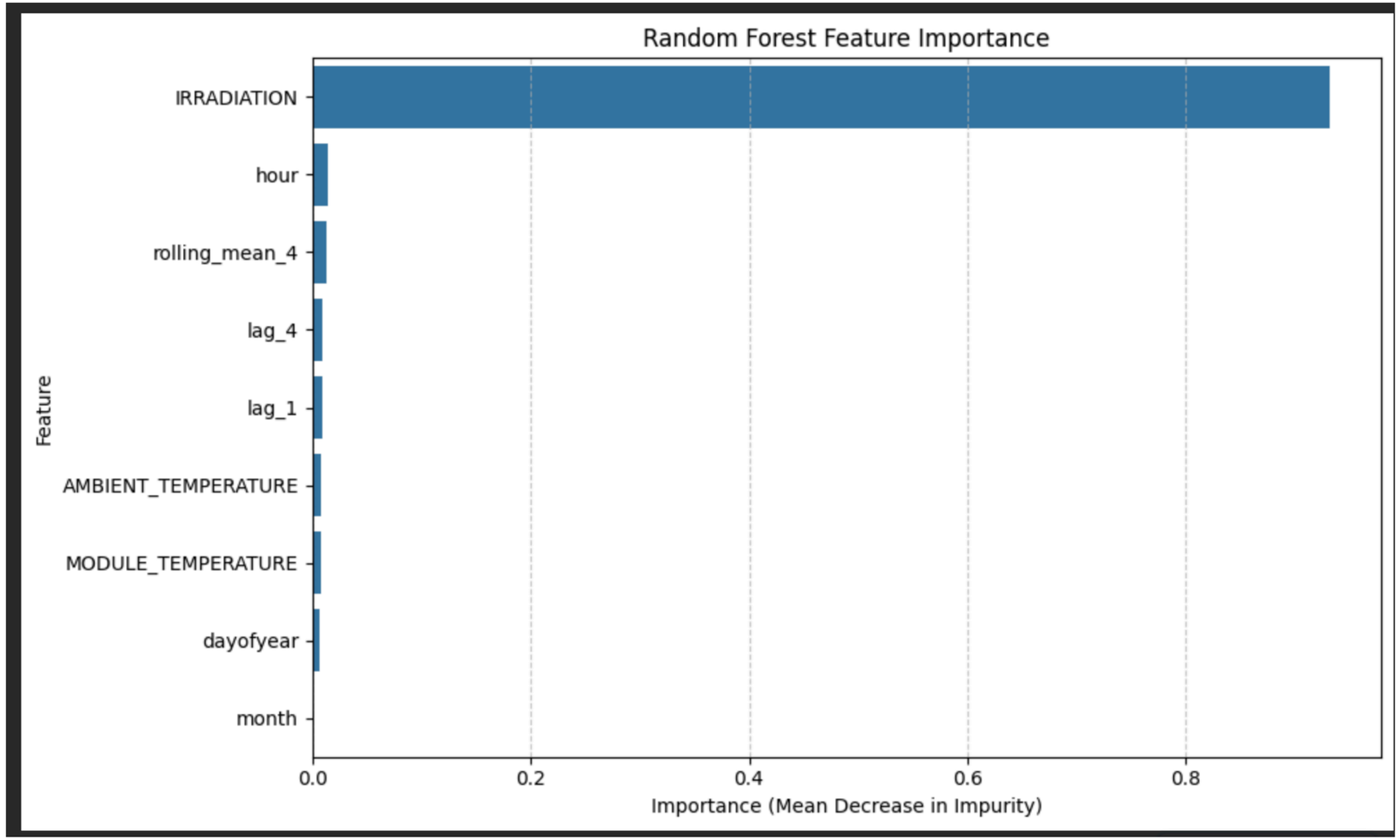
```python
import matplotlib.pyplot as plt

plt.figure(figsize=(12,5))
plt.plot(y_test.values[:200], label="Actual")
plt.plot(preds[:200], label="Predicted")
plt.legend()
plt.title("15-Minute Ahead Solar Forecast")
plt.show()
```



VISUALIZATION

Comparison of actual and predicted solar output

Understanding Feature Importance in Solar Forecasting



IMPORTANCE

Visual representation of feature influence on predictions.

Discussion & Insights

VARIANCE EXPLAINED

The model effectively explains **approximately 93%** of the variance in solar power generation, demonstrating its strong predictive capabilities and overall accuracy.

PERFORMANCE METRICS

Metrics such as MAE and RMSE indicate the model's typical absolute and squared errors, showcasing its reliability in forecasting solar energy.

PREDICTION STABILITY

Incorporating short-term lags in the feature set enhances prediction stability, contributing to more accurate solar power forecasts in dynamic weather conditions.

Deployment & Future Directions

STREAMLIT

The model has been successfully deployed using Streamlit, enabling **real-time inputs** for solar power prediction and enhancing user accessibility.

WEATHER APIS

Future improvements include integrating **live weather APIs** to refine forecasts, allowing enhanced accuracy in predicting solar energy generation.

ADVANCED TECHNIQUES

Exploring **cloud deployment** options and experimenting with advanced models like LSTM and XGBoost will further optimize forecasting accuracy and performance.

Team details

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